

Introduction & Hook

“Where does a VR system store all 3D objects, textures, and environment data?”

A VR system is not just rendering in real time.
It must store:

- 3D models
- Textures
- Object properties
- Physics data
- Lighting data
- User interactions

All of this is managed using a **VR database**.



PART 1: What is a VR Database?

Basic Concept

Definition:

A VR database is a structured collection of data used to represent objects, environments, and interactions in a virtual world.

It contains:

- Geometry data (vertices, meshes)
- Texture data
- Material properties
- Lighting information
- Object behaviors
- Physics attributes

Explain:

In traditional databases, we store:

- Text
- Numbers
- Records

In VR databases, we store:

- 3D objects
- Spatial relationships
- Scene hierarchy



PART 2: Scene Database

Scene Database Concept

Definition:

A scene database stores all objects and their relationships within a virtual environment.

It manages:

- Object position
 - Orientation
 - Scale
 - Parent–child relationships
 - Visibility
 - Collision properties
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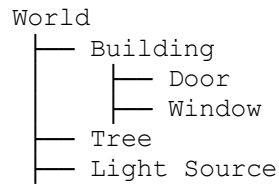
Structure of Scene Database (Scene Graph Concept)

most VR systems use a **Scene Graph**.

A scene graph:

- Is a tree structure
- Organizes objects hierarchically

Example:



If building moves:

→ Door and window move automatically (parent-child relation)

Why Scene Database is Important

- Efficient rendering
 - Easy object management
 - Faster culling
 - Better organization
 - Supports LOD and optimization
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PART 3: Static vs Dynamic Databases

Static Database

Definition:

A static database contains objects and data that do not change during execution.

Examples:

- Background mountains
- Buildings
- Terrain
- Skybox

Characteristics:

- Loaded once
- No modification

- Less computation required
 - Faster rendering
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Dynamic Database

Definition:

A dynamic database contains objects whose properties change during runtime.

Examples:

- Moving characters
- Interactive doors
- Physics objects
- Multiplayer avatars

Characteristics:

- Updated every frame
 - Requires real-time computation
 - Connected to physics and interaction modules
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Comparison Table

Feature	Static Database	Dynamic Database
Changes during runtime	No	Yes
Performance cost	Low	High
Examples	Buildings, terrain	Characters, vehicles
Requires updates	No	Yes

How Static & Dynamic Databases Work Together

In a VR factory simulation:

- Static:
 - o Machines

- o Walls
 - o Floor
- Dynamic:
 - o Worker avatar
 - o Moving robotic arm
 - o Conveyor belt items

Engine updates dynamic objects every frame, while static objects remain unchanged.

Recap + Questions

Recap Key Points:

- VR database stores all virtual world data
 - Scene database organizes objects hierarchically
 - Static database = fixed objects
 - Dynamic database = interactive objects
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Quick Oral Questions:

- What is a scene database?
- Give examples of static objects in VR.
- Why are dynamic databases more complex?
- How does scene graph help in rendering?